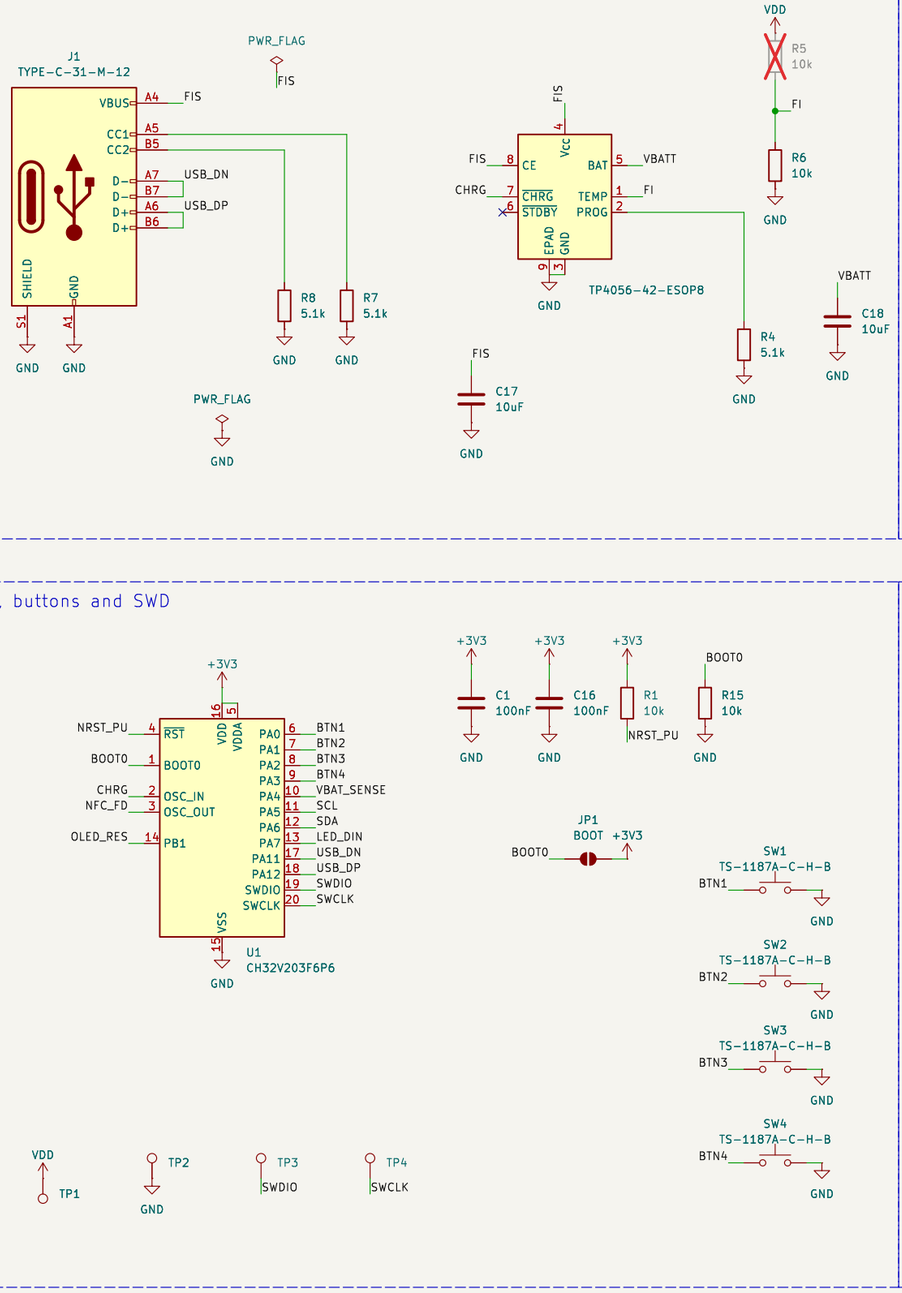
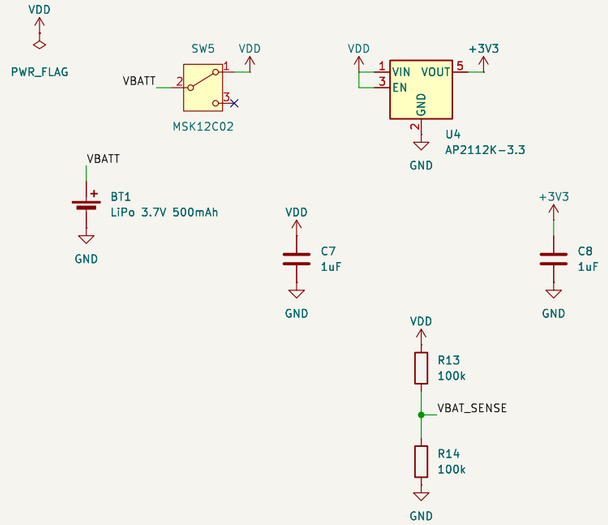
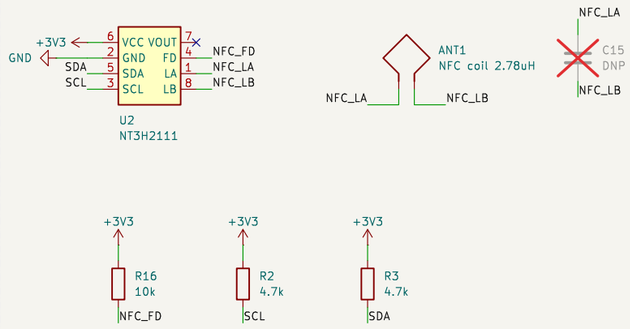
USB-C and charging



Battery, switch and 3V3

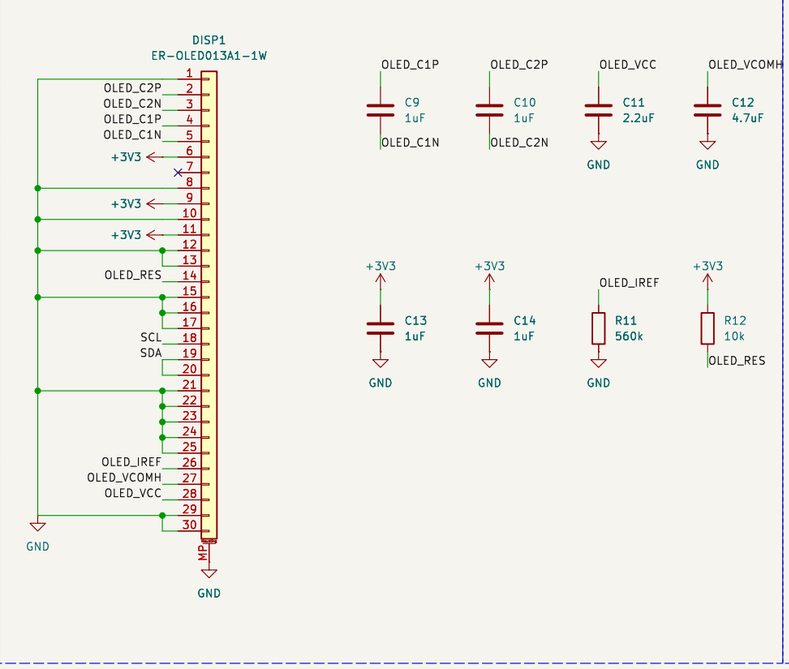


NFC tag

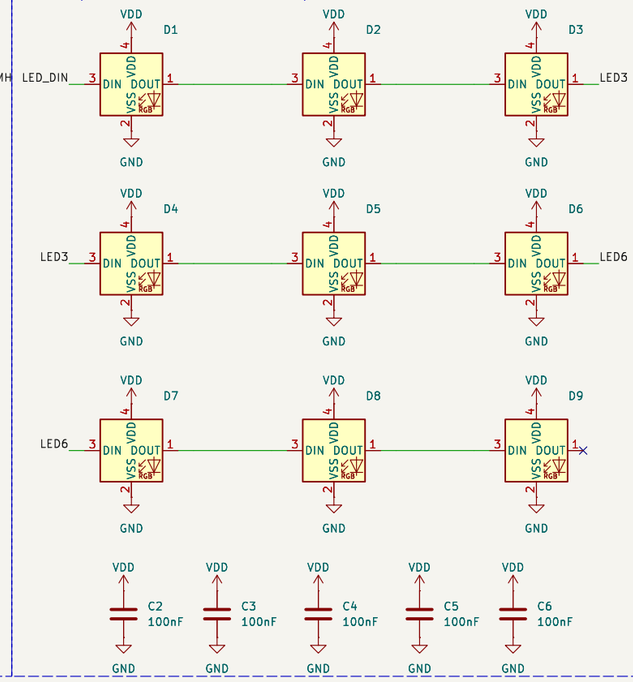


MCU, buttons and SWD

OLED



LEDs (D1–D9 WS2812B-2020)



Notes

3V3: AP2112K-3.3 LDO after the power switch, feeds the MCU, NFC chip, OLED logic and pull-ups.

NFC: 42 mm coil on B.Cu (2.78 uH) with the NT3H2111 50 pF tunes to 13.56 MHz. C15 is an optional tuning spot (DNP).

OLED: ER-OLED013A1-1W (SSD1306) in a FH12-30S ZIF, I2C mode (BS0=0 BS1=1 BS2=0, D/C#=0), address 0x3C.

MCU: PD0/PD1 used as GPIO for CHRG and NFC_FD, I2C bit-banged on PA5/PA6, LED data on PA7. Program over SWD or the USB bootloader.

Battery sense: VBAT_SENSE = VDD/2 on PA4 (ADC4).

Assembly: the LiPo sits on the back over the NFC coil, with a ~45 mm ferrite sheet between battery and board.

Sheet: /
File: technova_badge.kicad_sch

Title: TechNova badge

Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.8		Id: 1/1